REFERENCE • MHD-WAAM THESIS • caseMatrix.csv

The Case Matrix, Explained

Every parameter, every case, and which of them come from the paper

Sabbir Dipto | 31 cases, 38 columns | baseline: Zhao, Wei, Long, Chen, Liu & Ou (2021), Welding in the World 65:1571–1590

1. The short answer

DOES THE PAPER SIMULATE ALL 31 CASES? NO.

The paper simulates four configurations. Everything else in this matrix is either required to make the replication defensible as a thesis, or is your own contribution.

Of 31 rows: **5** reproduce the paper, **2** are the mesh-independence study the paper never reports, **4** measure numerical noise, **3** test parameters the paper had to guess, **3** feed ExaCA, and **13** are the magnetic-field study — which does not appear in the paper at all, because it is the point of your thesis.

What Zhao et al. (2021) actually simulate

Section 4 of the paper reports four deposition configurations, all at one set of process parameters (Table 2: 135 A, 16 V, 120 cm/min travel, 900 cm/min wire feed):

PAPER CONFIGURATION	MEANING	REPORTED IN	THIS MATRIX
Droplet transfer	Formation, necking and detachment at the wire tip	Figs. 4-6, Eq. 30	`D00`
SPSL	Single pass, single layer (bead on plate)	Figs. 7-11, Table 5	`R00`
SPTL, with idle time	Single pass, two layers, cooled to 300 K between	Figs. 12a-b, 13, Table 5	`R03`
SPTL, no idle time	Single pass, two layers, deposited back to back	Figs. 12c-d, 13, Table 5	`R04`
SPFL	Single pass, five layers, with idle time	Figs. 15-17, Table 5	`R05`

Five rows of this matrix map one-to-one onto the paper. The other 26 do not, and Section 2 explains why each exists.

A GAP THIS DOCUMENT FOUND

Writing this reference revealed that **R05-SPFL was missing** from the matrix. The five-layer case is the paper's *best-validated* configuration — Table 5 gives only 4.3% height error and 0.5% length error against experiment — so it is the strongest single validation gate available. It has now been added (disabled by default; it is the most expensive replication case).

2. Why 31 cases and not 5

A replication is not a thesis. The paper is a published result; reproducing it demonstrates that your model works, but contributes nothing new and would not survive an examiner asking what *you* did. The additional cases fall into five groups, each answering a question the paper does not.

GROUP	N	WHY IT EXISTS	IN PAPER?
replication + **paperfidelity**	5	Reproduce Zhao et al. Establishes that your solver is correct before you change anything. Validated against Table 5.	Yes
magnetic	13	The thesis. Field magnitude 0-100 mT, three orientations, three excitation frequencies. Zhao et al. 2022 measured these effects experimentally on the same alloy and rig but ran no simulation.	No
noisefloor	4	Measure the numerical resolution limit. Required because the solution is chaotic (see the companion note): without it, no magnetic effect can be attributed to physics rather than round-off.	No
meshstudy	2	Mesh independence at 0.30 and 0.40 mm against the 0.25 mm baseline. The paper reports no mesh study; a thesis must.	No
sensitivity	3	Test parameters the paper had to estimate: the Darcy constant, and the gas-conductivity floor your new electric-potential equation introduces.	No
thermal	3	Five-layer thermal histories in AdditiveFOAM to feed ExaCA. The microstructure leg of the thesis, which the paper does not attempt.	No

Enabled versus defined

Only 17 of the 31 are switched on. A row with `enabled=0` is not dead weight — it is a decision already made and documented, waiting for the phase that needs it. The mesh study belongs to Phase 3, the sensitivity runs to Phase 7, the ExaCA cases to Phase 6. Defining them now means the numerics are already frozen consistently when their turn comes.

3. Parameter reference

Thirty-eight columns, in ten groups. The **source** column is the important one: it separates what the paper specifies from what you introduced, which is exactly the boundary between replication and contribution.

3.1 Identity and control

Bookkeeping. None of these reach the solver.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
`case_id`	-	`unique`	yours	Folder name of the generated case. Prefix encodes purpose: D droplet, R replication, M magnetic, S sensitivity, T thermal.
`model`	-	`D / P / T`	yours	Which of the three models. D resolves the droplet, P the melt pool, T the multi-layer thermal field. Selects the template and the domain defaults.
`enabled`	-	`0 / 1`	yours	Whether the generator builds it. Lets a decision be recorded without being run.
`group`	-	`6 values`	yours	Audit unit. Numerics are checked for consistency within a (group, model) pair, because only cases in the same group are compared to each other.
`notes`	-	`free text`	yours	Why the case exists and what its acceptance gate is. Carried into case_meta.json.
`solver`	-	`waamMhdFoam / additiveFoam`	yours	Executable named in controlDict and in the generated Allrun.

3.2 Process parameters

Straight from Table 2 of the paper. Identical in every case — the experiment varies the magnetic field, not the welding schedule.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
`I_A`	A	`135`	Table 2	Deposition current. Drives arc power, the self-induced electromagnetic force (Eqs 16-19, 22-24), arc pressure (Eq 25) and, in your extension, the current density that produces the external Lorentz force.
`U_V`	V	`16`	Table 2	Deposition voltage. With current and arc efficiency it sets total heat input.
`travel_speed_m_s`	m/s	`0.02`	Table 2	Torch travel speed. 120 cm/min in the paper's units. Moves the Goldak source centre as x0 + v_T t.
`wire_feed_m_s`	m/s	`0.15`	Table 2	Wire feed rate. 900 cm/min. Becomes the inlet velocity of the liquid-metal inlet (assumption 1: wire feeding is modelled as hot liquid from a virtual tube).
`wire_dia_m`	m	`0.0012`	Table 2	ER5356 wire diameter. Halved to R_w, which sets the inlet radius and appears in the electromagnetic force expressions.
`gas_flow_lpm`	L/min	`20`	Table 2	Argon shielding gas flow. Feeds the annular inlet velocity profile of Eq. 28.

3.3 Magnetic field

Entirely new. Nothing here appears in the 2021 paper.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
`B_mT`	mT	`0, 10, 30, 50, 100`	yours	Applied flux density. The sweep brackets the crossover at roughly 20-30 mT where the Lorentz force overtakes Marangoni, established by scaling analysis.
`B_dir`	-	`x / y / z / none`	yours	Field orientation. Determines the force direction through J x B and therefore whether the half-domain symmetry survives. See 3.5.
`B_freq_Hz`	Hz	`0, 10, 50, 70`	yours	Excitation frequency. Zero is a static field. 70 Hz matches the alternating field used experimentally by Zhao et al. (2022) on this alloy.

3.4 Build strategy

From the paper's deposition configurations.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
`n_layers`	-	`1, 2, 5`	Section 4	Number of deposited layers. 1 = SPSL, 2 = SPTL, 5 = SPFL.
`idle_time_s`	s	`0 or 999`	Section 4.3	Interlayer cooling. 0 means the next layer starts immediately (the paper's case 2). 999 is a sentinel for 'enough time to reach 300 K', implemented as the paper does it: reset the domain temperature rather than simulate the wait.
`scan_mode`	-	`unidir / bidir`	Section 4.3	Deposition direction. The paper uses reciprocating (bidirectional) stacking for multi-layer builds, which avoids hump and collapse at the arc-on and arc-off ends.

3.5 Domain and mesh

Numerical choices. The symmetry entry is the one with a physical trap in it.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
`sym`	-	`half / full`	yours	Half domain exploits the XZ symmetry the paper uses. VALID ONLY when the force field is symmetric about y=0. With J mostly in -z: a y-directed field gives force along x and PRESERVES symmetry; x- or z-directed fields BREAK it. A longitudinal case run on a half domain produces a physically meaningless result that still renders plausibly.
`cell_size_m`	m	`0.0002 - 0.0004`	Sec. 3.7 / yours	Uniform cell size. The paper uses 0.25 mm minimum; the reduced cases use 0.35 mm, justified by the mesh study.
`deposit_len_m`	m	`0.008 - 0.040`	Sec. 2 / yours	Deposit length. The paper deposits 40 mm; reduced cases use 20 mm, justified by the pool reaching a quasi-steady width and height well before that.

3.6 Time control

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
dt_s	s	4e-5 - 1e-3	Sec. 3.7 / yours	Initial and maximum time step. The paper's 4e-5 s sits inside the capillary limit sqrt(rho dx^3 / 2 pi gamma), about 9e-5 s at 0.25 mm. Model T is conduction-dominated and tolerates 1e-3 s.
end_time_s	s	0.1 - 12	derived	Physical duration. Left blank, the generator computes length / speed x layers.
write_interval_s	s	0.001 - 0.05	yours	How often fields are written. The dominant control on disk usage; combined with binary format and compression in controlDict.
purge_write	-	0	yours	Keep only the last N time directories. Zero keeps everything; set it during exploratory runs to protect disk.

3.7 Heat source

Goldak double ellipsoid, Eq. 27. Calibration constants, not physical measurements.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
goldak_af_m	m	0.003	Sec. 3.5	Front semi-axis. Shorter than the rear: the moving arc leaves a wide trailing lobe and a narrow leading one.
goldak_ar_m	m	0.006	Sec. 3.5	Rear semi-axis.
goldak_b_m	m	0.004	Sec. 3.5	Transverse semi-axis (y).
goldak_c_m	m	0.004	Sec. 3.5	Depth semi-axis (z).
eta_arc	-	0.70	Table 4	Arc efficiency for heating the pool, excluding the fraction that melts the wire. Arc power delivered = eta x U x I = 1512 W.

3.8 Arc and current distribution

Both are estimates in the paper, which is why they get sensitivity runs.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
sigma_j_m	m	0.002	Table 4	Gaussian distribution radius of the current density. Marked 'estimated' in Table 4. Sets the current concentration factor and the top-surface boundary condition of your new electric-potential equation.
sigma_p_m	m	0.002	Table 4	Distribution radius of arc pressure, Eq. 25. Also marked 'estimated'.

3.9 Model constants

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
A_darcy	kg/(m^3 s)	1e6 - 1e7	Sec. 3.2	Mushy-zone constant in the Darcy momentum sink, Eq. 7. Higher values arrest solidifying metal faster. The paper notes 1e4-1e7 as the usual range and that too large a value makes the solution oscillate. S02 tests 1e7 against the known bead-widening artefact.
sigma_e_gas_ratio	-	1e-2 - 1e-4	yours	Floor on argon's electrical conductivity as a fraction of the metal's. Argon is effectively an insulator, but a conductivity ratio beyond about 1e6 makes the potential matrix so ill-conditioned that the linear solver stalls. S00 and S01 demonstrate that the pool solution is insensitive to this floor across a decade.

3.10 Execution and verification

Added after the Phase 0 chaotic-sensitivity finding.

COLUMN	UNIT	VALUES	SOURCE	WHAT IT DOES
n_procs	-	2 or 6	yours	MPI ranks. Six physical cores; the other six threads are SMT and CFD is memory-bandwidth-bound, so 12 ranks would run slower. Model D uses 2 because 48k cells split six ways spends more on communication than it saves. Written into decomposeParDict by the generator.
decomp_method	-	scotch	yours	Decomposition algorithm. Scotch partitions by graph connectivity, so it handles graded meshes and directions only one cell thick, where 'simple' fails outright.
p_rgh_tol	-	1e-07 / 1e-08	yours	Pressure linear-solver tolerance. The mechanism of the noise-floor protocol: a twin case differs from its reference ONLY in this value, a change with no physical content, and the resulting spread is the numerical resolution limit.
noise_twin_of	-	case_id or blank	yours	Marks a case as the noise twin of another. The generator audits that a twin differs from its reference in nothing but p_rgh_tol, and refuses to build if it has drifted.

3.11 Values the generator derives

These are not columns. The CSV holds intent; the generator holds arithmetic, so a reader sees decisions rather than derived clutter. All are available as tokens and land in constant/caseParameters.

DERIVED	FROM	WHY
Rw_m	wire_dia_m / 2	Wire radius, used by the inlet BC and the EM force expressions
Bx_T, By_T, Bz_T	B_mT and B_dir	Field resolved onto axes, in tesla, as the vector Bext
B_omega_rad_s	2 pi f	Angular frequency for an alternating field
x_min_m .. z_max_m	model, deposit_len_m, sym	Domain box. Model D is centred on the wire tip; P and T span the deposit
nx, ny, nz, n_cells	box / cell_size_m	Uniform mesh counts and the cell-count upper bound
end_time_s	length / speed x layers	Only when the column is left blank
arc_power_W	eta x U x I	1512 W at the baseline settings

4. Every case

In build order. The **on** column marks whether it is currently enabled; the **phase** column is when the roadmap calls for it.

CASE	ON	PH	MODEL	WHAT IT IS FOR
`D00-droplet-135A`	yes	2	D	Resolves droplet formation, necking and detachment at the wire tip. Reproduces the paper's Figs. 4-6. Gate: radius within ~5% of the 1.105 mm predicted by Eq. 30, detachment near 30 Hz, peak internal velocity 0.9 m/s. Validates surface tension, electromagnetic force and mass input all at once, on a domain small enough to iterate on.
`D00-droplet-135A-noise`	yes	2	D	Noise twin of D00. Establishes the resolution limit on droplet radius, frequency and peak velocity before those numbers are quoted.
`D01-droplet-110A`	-	-	D	Lower current. Checks that transfer mode and droplet size respond sensibly to current. Not in the paper, which reports one current only.
`D02-droplet-160A`	-	-	D	Higher current. Same purpose; together with D01 it brackets the globular transfer regime.
`R00-SPSL-paper`	yes	3	P	The paper's SPSL configuration at the paper's own resolution: 0.25 mm cells, 40 mm deposit, 2 s. The primary replication gate. Table 5 gives experiment 1.80 mm height against simulation 1.55 mm; you should land in the same territory. This is the safety point of the whole project.
`R00-SPSL-paper-noise`	-	3	P	Noise twin at paper fidelity. Disabled by default because it doubles the most expensive replication run; enable once the baseline converges.
`R01-SPSL-mesh030`	-	3	P	Mesh independence at 0.30 mm. The paper reports no mesh study; a thesis must have one.
`R02-SPSL-mesh040`	-	3	P	Mesh independence at 0.40 mm. With R00 and R01 this justifies the 0.35 mm used for the magnetic matrix.
`R03-SPTL-idle`	-	4	P	Two layers with interlayer idle time, bidirectional. The paper's case 1. Cooling is implemented as the paper does it, by resetting the domain to 300 K.
`R04-SPTL-noidle`	-	4	P	Two layers deposited back to back. The paper's case 2. Together with R03 it reproduces Fig. 13: idle time gives a taller, narrower bead.
`R05-SPFL`	-	4	P	Five layers with idle time. The paper's best-validated configuration: Table 5 reports only 4.3% height error and 0.5% length error. Strongest available validation gate, and the most expensive replication case.
`M00-B000`	yes	5	P	Zero-field baseline at reduced resolution. Every magnetic case is measured against this one, so it must reproduce R00's trends before any of them mean anything.
`M00-B000-noise`	yes	5	P	Noise twin of the magnetic baseline. Run this BEFORE interpreting any magnetic case; it defines what counts as a result.
`M01-By-010`	yes	5	P	Transverse-Y at 10 mT. Below the estimated Marangoni crossover, so it tests whether a weak field perturbs the pool measurably at all.
`M02-By-030`	yes	5	P	Transverse-Y at 30 mT, close to where the Lorentz force is predicted to overtake Marangoni.
`M03-By-050`	yes	5	P	Transverse-Y at 50 mT. Comfortably above the crossover.
`M04-By-100`	yes	5	P	Transverse-Y at 100 mT, where the applied Lorentz force is comparable to the self-induced one.
`M05-Bx-030`	yes	5	P	Transverse-X at 30 mT. Force acts along y, deflecting the pool sideways and producing bead asymmetry. Full domain required.

CASE	ON	PH	MODEL	WHAT IT IS FOR
M06-Bx-050	yes	5	P	Transverse-X at 50 mT. Full domain.
M07-Bz-030	yes	5	P	Longitudinal at 30 mT. Azimuthal force: classical electromagnetic stirring, the configuration used experimentally for grain refinement. Full domain.
M08-Bz-050	yes	5	P	Longitudinal at 50 mT. Full domain.
M09-Bz-100	yes	5	P	Longitudinal at 100 mT. Full domain.
M10-Bz050-f010	yes	5	P	Alternating longitudinal, 50 mT at 10 Hz. Slow enough that the pool largely follows the field.
M11-Bz050-f050	yes	5	P	Alternating at 50 Hz. Approaching the pool's own response time.
M12-Bz050-f070	yes	5	P	Alternating at 70 Hz. Matches the excitation frequency Zhao et al. (2022) found optimal on this alloy and rig, which makes it the case with a direct experimental counterpart.
S00-sigmafloor-1e2	-	7	P	Gas conductivity floored at 1e-2 of the metal. Upper end of the sensitivity study on a parameter your electric-potential equation introduces.
S01-sigmafloor-1e4	-	7	P	Floor at 1e-4. With S00 and the 1e-3 baseline this demonstrates insensitivity across a decade, pre-empting the obvious examiner question.
S02-darcy-1e7	-	7	P	Darcy constant raised to 1e7 to suppress the post-solidification bead creep that the paper itself reports as a limitation of the enthalpy-porosity method.
T00-B000-5layer	-	6	T	Five-layer thermal history in AdditiveFOAM with geometry frozen from Model P. Produces the time-temperature data ExaCA consumes. Zero field.
T00-B000-5layer-noise	-	6	T	Noise twin of the thermal baseline, so the extracted G and R carry a resolution limit.
T08-Bz050-5layer	-	6	T	Five-layer thermal history with a 50 mT longitudinal field. Paired with T00, this is the comparison that drives the microstructure prediction.

5. Using it

- `python3 generate_cases.py --list` — every row, with disabled ones marked.
- `--dry-run` — report cell counts and end times, write nothing.
- `--pairs` — the noise-floor pairs and their perturbations.
- `--only "M0*"` or `--group magnetic` — build a subset.
- `--force` — overwrite cases that already hold run output. Without it, the generator refuses.

The CSV is the single source of truth. Never edit a generated case folder: regenerating silently discards the change. Edit the CSV and rerun, or edit the template if the change applies to every case.

> **WHAT THE AUDIT ENFORCES**
>
> The generator refuses to build when the frozen numerics (n_procs, decomp_method, cell_size_m, dt_s) differ within a (group, model) pair, or when a noise twin differs from its reference in anything but p_rgh_tol. Groups that deliberately vary numerics — meshstudy, sensitivity — are exempt. A comparison group with no enabled noise twin raises a warning rather than an error.